Date

Residue

$$X(z) = \frac{z^{-1}}{1 - 0.25z^{-1} - 0.375z^{-2}} \times \frac{z^2}{z^2}$$

① $X(n) \longrightarrow 4$ sumol $\longrightarrow$ Residue

① $X(z)$ [illegible]

$$\boxed{X(z) = \frac{z}{z^2 - 0.25z - 0.375}}$$

$$F(z) = X(z) \cdot z^{n-1}$$

$$F(z) = \frac{z^1}{z^2 - 0.25z - 0.375} \times z^{n-1}$$

$$= \frac{z^n}{z^2 - 0.25z - 0.375} = \frac{z}{(z+0.5)(z-0.75)}$$

$n = 1$

$$\frac{-4}{5}(-0.5) + \frac{4}{5}(0.75)$$

$$\frac{4}{10} + \frac{12}{20}$$

$$= \frac{4}{10} + \frac{6}{10} = 1$$

$$X(n) = \text{Res}_1 \Big[F(z), (z+0.5) \Big]_{z=-0.5} + \text{Res}_2 \Big[F(z), (z-0.75) \Big]_{z=0.75}$$

$$X(n) = \frac{(-0.5)^n}{-1.25} = \frac{-4}{5}(-0.5)^n + \frac{4}{5}(0.75)^n$$

$X(0) = 0$

ElSalam

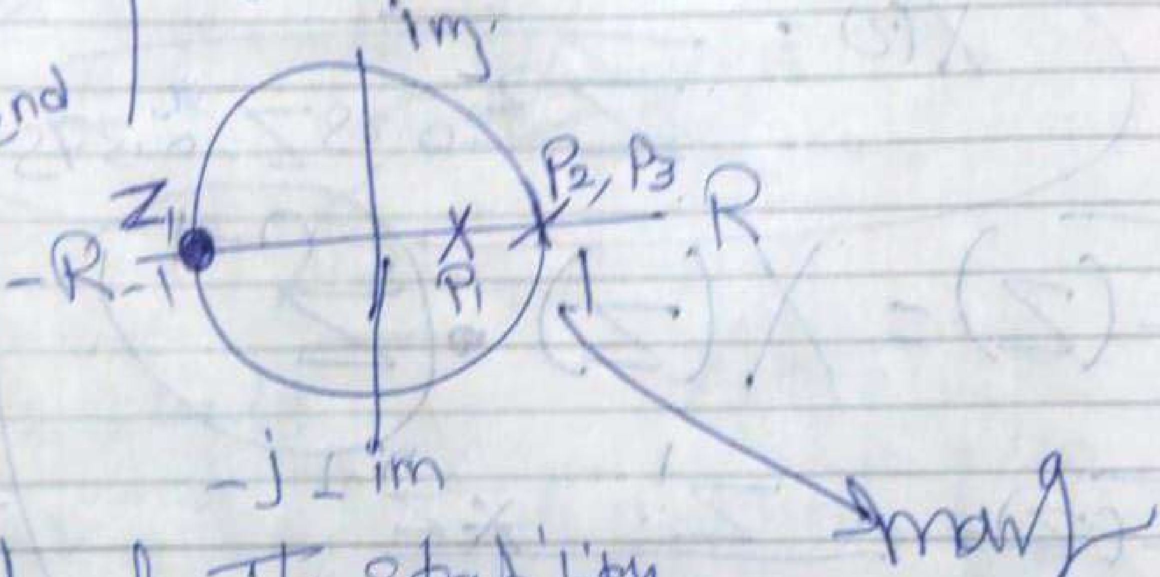
$$H(z) = \frac{z+1}{(z-0.5)(z-1)^2}$$

① Draw Pole Zero diagram

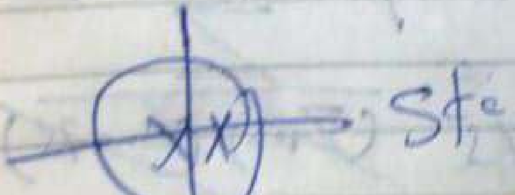
Poles X | Zero O

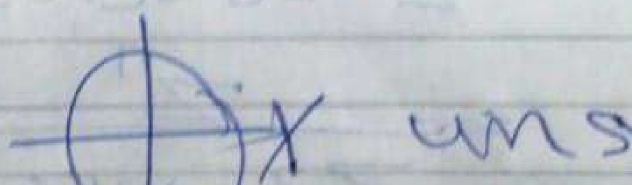
$Z = 0.5$ | $Z = -1$

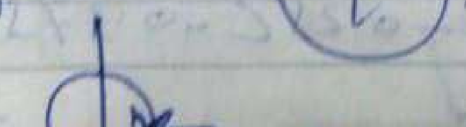
$Z = 1$ ' end



② check The stability







③ $ROC = Z \{0.5, 1\}$ ag

Elsalam

$h(n)$? $P.F \longrightarrow 4$

$$\boxed{\frac{H(z)}{z}}$$

P.F

$$\frac{H(z)}{z} = \frac{(z+1)}{(z-0.5)(z-1)^2} \times \frac{1}{z}$$

$$= \frac{(z+1)}{z(z-0.5)(z-1)^2 \cancel{(z-1)}}$$

$$= \frac{A}{z} + \frac{B}{z-0.5} + \frac{C}{\cancel{(z-1)}} + \frac{D}{(z-1)^2}$$

$$A = \frac{1}{(-0.5)(-1)^2} = \boxed{-2}$$

$$B = \frac{1.5}{0.5 \times 0.25} = 1.5 \cdot 2 \cdot 4 = \boxed{12}$$

$$D = \frac{2}{0.5} = 4$$

$$C = \frac{d}{dz} \frac{z+1}{z(z-0.5)} = \frac{z+1}{z^2-0.5z} = \frac{(z^2-0.5z) - (2z-0.5)(z+1)}{(z^2-0.5z)^2}$$

$$= 4(0.5 - 1.5 * 2) = 4 \times -2.5 = -10$$

S.Salam

Date: No.

ch#4 مراجعة

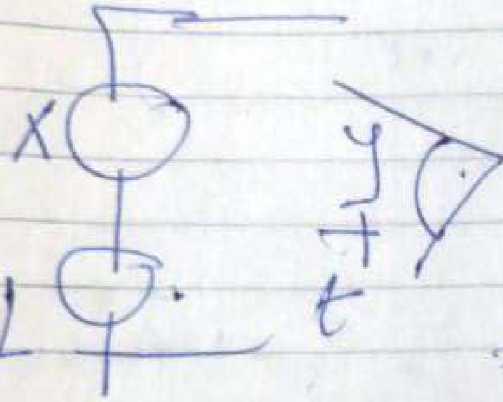
Define: linearsystem, Non line, Causal, NonCau
BiBo

$x \longrightarrow y$

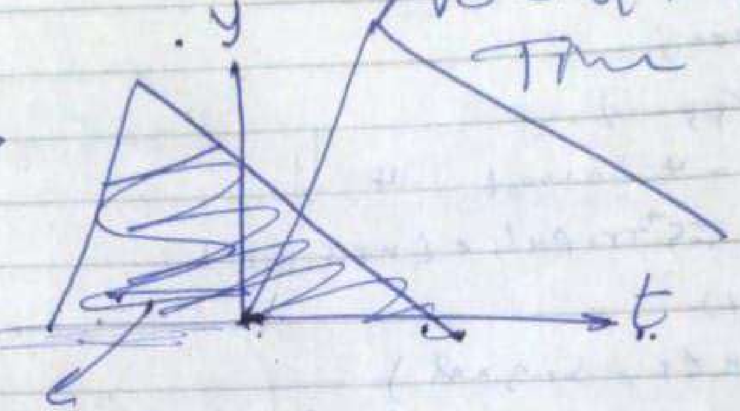
$L \longrightarrow t$

$x+L \longrightarrow y+t$

memory



No output befor Time



No Causal.

Causal
Relusively

$$X(0) = \frac{a_0}{b_0}$$

$$X(n) = \frac{a_n - \sum x(n-i)\, b_i}{b_0}$$

↳ a_n

$X(n) \longrightarrow X(z)$ X

$X(z) \longrightarrow X(n)$

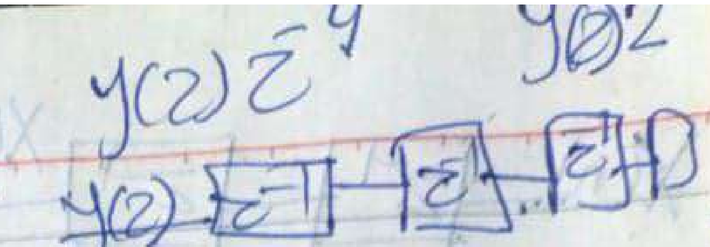
→ Pow X long wk

→ P.F.

→ Res

Date: No

Sheet = 5



Draw the Realization Structure using Block diagram!

$$H(z) = \frac{z+1}{z^3 - 4z^2 + 3z - 1} \times \frac{z^{-3}}{z^{-3}}$$

Difference equation

Negative

$$\frac{y(z)}{x(z)} = H(z) = \frac{z^{-2} + z^{-3}}{1 - 4z^{-1} + 3z^{-2} - z^{-3}}$$

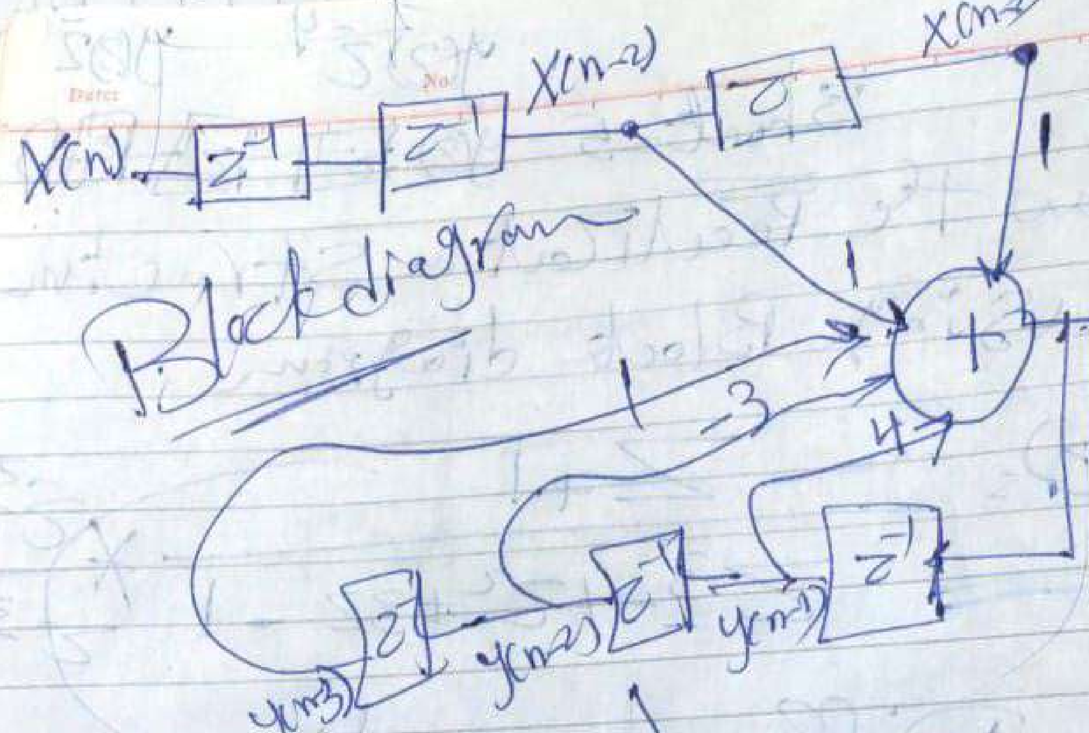
$$y(z)\left[1 - 4z^{-1} + 3z^{-2} - z^{-3}\right] = X(z)\left[z^{-2} + z^{-3}\right]$$

$$y(z) - 4y(z)z^{-1} + 3y(z)z^{-2} - y(z)z^{-3} = X(z)z^{-2} + X(z)z^{-3}$$

$$y(n) - 4y(n-1) + 3y(n-2) - y(n-3) = X(n-2) + X(n-3)$$

Elsalam

$$y(n) = 0X(n-2) + X(n-3) + 4y(n-1) - 3y(n-2) + y(n-3)$$



Blockdiagram

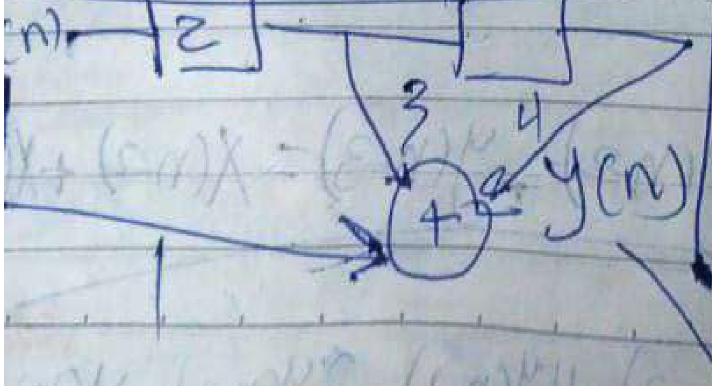
FIR

$$\frac{y(z)}{x(z)} = H(z) = 1 + 3z^{-1} + 4z^{-2}$$

$$y(z) = [X(z)[1 + 3z^{-1} + 4z^{-2}]$$

$$= X(z) + 3X(z)z^{-1} + 4X(z)z^{-2}$$

$$y(n) = x(n) + 3x(n-1) + 4x(n-2)$$



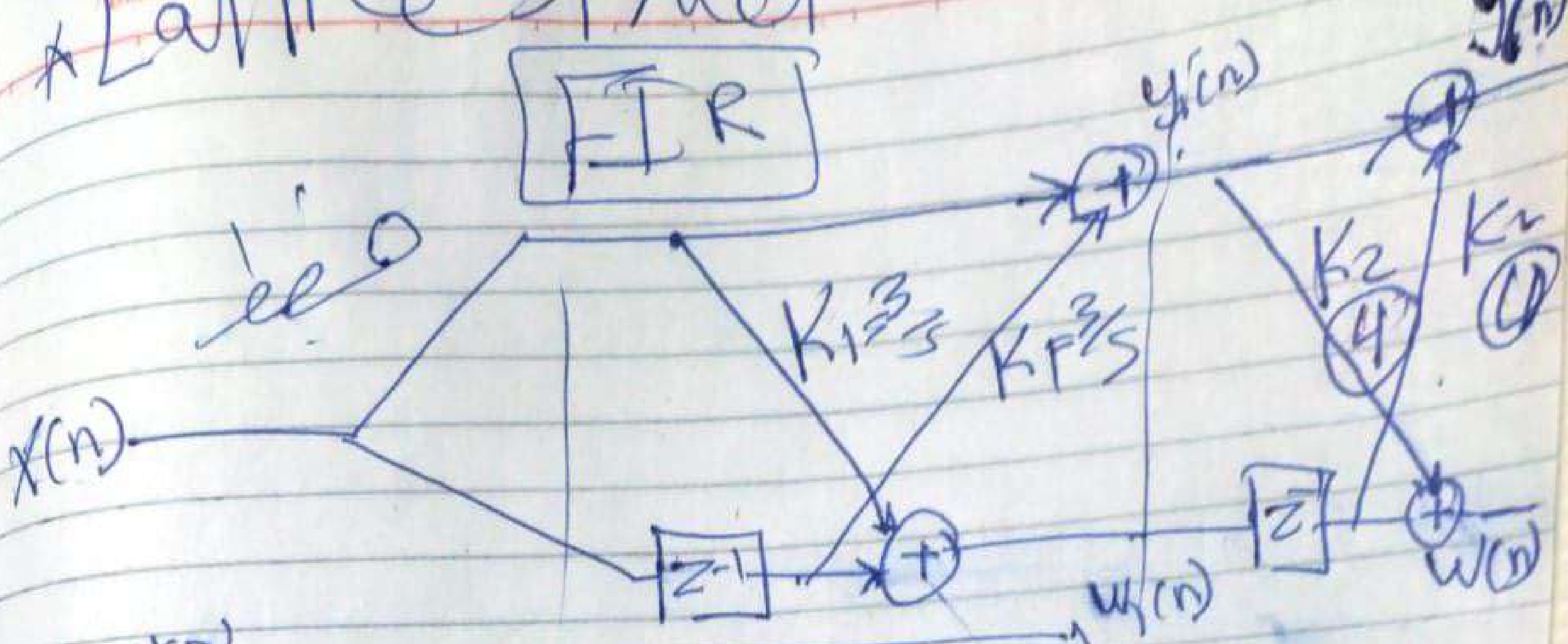
IIR

$$H(z) = \frac{1 + 3z^{-1} + 4z^{-2}}{1 + 5z^{-1} + 6z^{-2}}$$

Block d

Direct method

* Lattice structure



$$\cancel{H(z)}\ \frac{Y(z)}{X(z)} = h(0) + h(1) z^{-1} + h(2) z^{-2}$$

$= h(0)x(n) + h_1 x(n-1) + h_2 x_n$

$y(2) = x(2)h_0 + x(1)h(1)$

$y_1(n) = x(n) + K_1 x(n-1)$

$w_1(n) = K_1 x(n) + x(n-1) \longrightarrow w_1(n-1) = K_1 x(n-1) + x(n-2)$

$$y(n) = y_1(n) + K_2 w_1(n-1)$$

$= x(n) + K_1 x(n-1) + K_2 [K_1 x(n-1) + x(n-2)]$

$$\boxed{y(n) = x(n) + K_1 x(n-1) + K_1 K_2 x(n-1) + K_2 x(n-2)}$$

$= x(n) + (K_1 + K_1 K_2) x(n-1) + K_2 x(n-2)$

$y(n) = h(0)x(n) + h(1)x(n-1) + h(2)x(n-2)$

$h(2) = K_2$ | $K_1 + K_1K_2 = h(1)$, $K_1 = \frac{h(1)}{1+K_2}$

$K_1(1+K_2) = h(1)$, $K_1 = \frac{h(1)}{}$

Date:

$$H(z) = \frac{-2z}{z} + \frac{12z}{z-0.5} + \frac{-10z}{(z-1)} + \boxed{\frac{4z}{(z-1)^2}}$$

$H(z)$	$h(n)$
$\frac{Kz}{z-\alpha}$	$K\alpha^n$
$\frac{K\alpha z}{(z-\alpha)^2}$	$Kn\alpha^n$
K	$K\delta(n)$

$\frac{Kz}{z-1} \longrightarrow K$

$\longrightarrow Kn$

$K\alpha = 4 \quad \alpha = 1$

$k = \frac{4}{1}$

$$h(n) = -2\delta(n) + 12(0.5)^n - 10 + 4n$$

$$h(0) = -2\overset{1}{\boxed{\delta(0)}} + 12 - 10 = \text{zero}$$

$$h(1) = \cancel{-2\delta(1)} + \overset{6}{12 \times 0.5} \overset{-10+4}{- 10 + 4} = \text{zer}$$

$h(2)$

$h(3)$

$Y(z) \longrightarrow y(n)$

$Y(z)z^{-1} \longrightarrow y(n-1$

$Y(z)z^{-3} \longrightarrow y(n-3)$

$\boxed{z^{-1}}$

$\frac{3z}{z+5} \longrightarrow 3$

$\frac{3z}{}$

$\frac{3z}{z-(-5)}$